

Supplementary Material for “DriveToken: Learning Dynamic Vision-Token Budgets from Driving Rewards for Autonomous VLAs”

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1 Reproducibility Summary

This supplement provides architecture specifications, full training details, and extended results referenced in the main paper. All numbers are derived from the AutoVLA-3B backbone on NAVSIM navtest ($N=11,576$ scenes) unless stated otherwise. The confidence-based runtime fallback is an integral part of the DriveToken deployment pipeline and is reported explicitly throughout.

2 Architecture Details

The token scorer f_θ is a three-layer MLP:

Component	Specification
Input dimension	2051 = 2048 (ViT-to-LLM embedding) + 3 (camera one-hot)
Layer 1	LayerNorm(2051) $\rightarrow$ Linear(2051, 256) $\rightarrow$ GELU
Layer 2	Linear(256, 256) $\rightarrow$ GELU
Layer 3	Linear(256, 1)
Output	Scalar importance score per token
Total parameters	0.59M
Input feature	Standardized layer-0 ViT-to-LLM projection
Camera encoding	3-dimensional one-hot (front, left, right)
Inference cost	< 1 ms per scene (< 0.02% of total VLA FLOPs)

The budget head g_ϕ (for Stage 2 Budget RL):

Component	Specification
Input	Mean-pooled token features, $\bar{\mathbf{x}}_s \in \mathbb{R}^{2051}$
Layer 1	LayerNorm(2051) $\rightarrow$ Linear(2051, 256) $\rightarrow$ GELU
Layer 2	Linear(256, 128) $\rightarrow$ GELU
Layer 3	Linear(128, 1)
Output	Raw logit u_s ; squashed to $k_s \in [0.2, 0.9]$ via $k_s = 0.2 + 0.7 \cdot \text{sigmoid}(u_s)$
Total parameters	$\sim 0.01\text{M}$ (budget head only)

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3 Training Details

Item	Value
<i>Stage 1: LambdaRank scorer</i>	
Training data	4,000 navtrain scenes (80/20 train/val split)
Optimizer	AdamW, lr = 3×10^{-4} , weight decay 10^{-4}
Schedule	Cosine decay to 10^{-5}
Batch size	64 scenes per batch, 1,024 token pairs per scene
Max epochs	20; early stopping patience 3 on validation pairwise accuracy
<i>Stage 2: Budget RL (joint scorer + budget head)</i>	
Init	Stage 1 LambdaRank scorer
Epochs	5
Checkpoint selected	Epoch 4, step 2,180 (by PDMS on 500-scene validation subset)
Optimizer	AdamW
Scorer learning rate	1×10^{-5}
Budget head learning rate	1×10^{-4}
Warmup steps	100
Batch size	32 scenes per step
Group size G	8 scenes per group
Budget samples K per scene	4
Policy std init	$\log \sigma_0 = -1.0$
Reward weights	$\lambda_d=2.0, \lambda_s=0.5, \lambda_e=0.15$
Scorer trust-region β_{ref}	0.01 (weight-space L_2 anchor to SFT)
Budget trust-region	None (budget head learns freely)
Keep-ratio range	[0.2, 0.9]
Gradient sync	8 workers, synchronized DDP

Reward decomposition. The driving reward R_{drive} weights the six NAVSIM sub-metrics with $w=(0.35, 0.20, 0.15, 0.15, 0.10, 0.05)$ for (progress, collision, drivable, TTC, direction, comfort) and mixes absolute scores with per-scene pruning deltas with $(\alpha, \beta) = (0.3, 0.7)$. The safety penalty L_{safe} penalizes harmful degradation on (collision, drivable, TTC) with weights (0.4, 0.3, 0.3).

Training compute. Stage 1: ~ 4 GPU-hours on $1 \times \text{H20}$. Stage 2: ~ 240 GPU-hours on $8 \times \text{H20}$ (5 epochs $\times \sim 6$ hours/epoch). Each Stage 2 rollout requires one VLA forward pass per sampled budget action; with $G=8, K=4$ this

Method	PDMS	NC	DAC	EP	TTC	Comfort	Direction
No Prune ($r=1.0$)	0.8988	0.994	0.960	0.833	0.977	0.999	0.981
<i>Retain 360 tokens ($r=0.5$, 33.6% FLOPs reduction)</i>							
SFT Scorer (ours, +fallback)	0.9045	0.987	0.956	0.809	0.971	0.998	0.979
<i>Dynamic (scene-adaptive)</i>							
SFT + τ -cut ($\sim 60\%$ keep)	0.8949	0.994	—	0.829	—	—	—
Budget RL (35.5% keep, +fallback)	0.9107	0.994	—	0.842	—	—	—

Table 1: PDMS sub-metric breakdown on NAVSIM navtest ($N=11,576$). NC = no-at-fault collisions; DAC = drivable area compliance; EP = ego progress; TTC = time-to-collision.

Keep ratio bin	Scenes	%
[0.20, 0.30)	5,176	44.7%
[0.30, 0.40)	1,717	14.8%
[0.40, 0.50)	2,290	19.8%
[0.50, 0.62]	2,393	20.7%
Total	11,576	100%
Scenes < 40% retention	6,893	59.5%

Table 2: Per-scene keep ratio distribution of Budget RL on navtest ($N=11,576$).

yields 32 rollouts per training step. Hardware: NVIDIA H20 (97 GB VRAM).

4 PDMS Sub-Metric Breakdown

Table 1 reports the full six-submetric breakdown for key configurations on the full navtest split ($N=11,576$, physical token removal).

Key observation: At $r=0.5$, the SFT scorer with fallback achieves PDMS 0.9045, exceeding the unpruned baseline (0.8988). Budget RL further improves collision avoidance to 0.994 while reducing mean token retention to 35.5%.

5 Dynamic Keep Ratio Distribution

Table 2 reports the binned distribution of per-scene keep ratios produced by the Stage 2 Budget RL policy on the full navtest split. The budget head allocates keep ratios in $[0.20, 0.62]$ with mean 0.355 and standard deviation 0.12.

The budget head strongly skews toward aggressive pruning: 59.5% of scenes retain fewer than 40% of tokens, confirming that the learned policy discovers substantial redundancy in most driving scenes.

6 τ -cut Definition and Calibration

The τ -cut method provides a training-free scene-adaptive pruning baseline using the Stage 1 SFT scorer. For each scene, we retain all tokens whose score exceeds a global threshold:

$$\mathcal{S}_s = \{i \mid f_\theta(\mathbf{x}_i) > \tau\}, \quad K_s = |\mathcal{S}_s|. \quad (1)$$

The threshold τ is calibrated once on a held-out validation subset of navtrain:

Tag	Target mean keep ratio	Calibrated τ
kr040	40%	−0.1253
kr050	50%	−0.1487
kr060	60%	−0.1668
kr070	70%	−0.1840

The main paper reports kr060 (mean keep $\sim 60\%$, PDMS 0.8949). Budget RL (Section 3) learns a scene-conditional budget from reward feedback and achieves substantially lower mean retention (35.5%) at higher PDMS (0.9107).

7 Confidence-Based Runtime Fallback

To guard against overly aggressive pruning on high-uncertainty scenes, we deploy a confidence-based runtime fallback that operates *before* generation.

Confidence score. For a scene with $N=720$ visual tokens and scorer logits $\{f_\theta(\mathbf{x}_i)\}_{i=1}^N$, we compute the normalized mean entropy:

$$H_s = -\frac{1}{\log N} \sum_{i=1}^N p_i \log p_i, \quad p_i = \frac{\exp(f_\theta(\mathbf{x}_i))}{\sum_{j=1}^N \exp(f_\theta(\mathbf{x}_j))}. \quad (2)$$

$H_s \in [0, 1]$ measures the scorer’s uncertainty: $H_s \approx 1$ means near-uniform scores (high uncertainty), while $H_s \approx 0$ means the scorer is confident about which tokens matter.

Fallback protocol. If H_s exceeds a calibrated threshold τ_{conf} , the scene is classified as high-uncertainty and the system falls back to full 720-token inference. The threshold is calibrated on the navtrain validation set to yield a fallback rate of approximately 1–2%. The fallback is an integral part of the deployment protocol and is reported explicitly in all “ours” rows of the main table.

8 FLOPs Estimation

We estimate theoretical FLOPs for the AutoVLA-3B pipeline (Qwen2.5-VL-3B backbone). The ViT always processes all 720 vision tokens. Pruning occurs *after* the ViT, so FLOPs savings are entirely in the LLM prefill (36 layers, hidden 2048, FFN 11008, 16 heads with GQA 2 KV heads).

Table of FLOPs by retention ratio:

Retention	Kept	Seq.	ViT	LLM	Total	Saving
100%	720	941	949.6	5482.8	6433.3	—
75%	540	761	949.6	4393.6	5344.1	19.9%
50%	360	581	949.6	3323.6	4274.0	39.4%
35.5% (ours)	256	~477	949.6	~2698	~3648	~50.8%
25%	180	401	949.6	2272.6	3223.1	58.6%

9 Inference Latency Profile

Configuration	Latency (H20, AutoVLA-3B)
No prune ($r=1.0$)	2.16 s per scene
Pruned ($r=0.5$, physical drop)	2.08 s per scene (3.6% wall-clock reduction)
Scorer overhead	< 1 ms (< 0.05% of inference time)

Note on wall-clock vs. FLOPs. The 3.6% wall-clock reduction at $r=0.5$ is limited by the small baseline model size and fixed I/O overhead. The theoretical FLOPs reduction of 33.6% at $r=0.5$ and 43.3% in the dynamic configuration translates to larger absolute savings on larger-scale models where prefill cost dominates.